

Lab Experiment 16

Feed Forward Neural Network

Aim

To write a Python program to implement a Feed Forward Neural Network for classification.

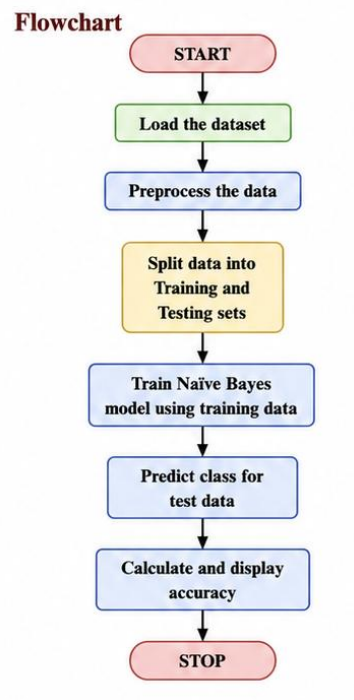
Objective

The objective of this experiment is to implement a simple Feed Forward Neural Network using Python. It aims to train the network and predict the output for given input data.

Algorithm

1. Initialize the input, hidden, and output layers.
2. Assign random weights and biases.
3. Pass the input through the hidden layer.
4. Apply an activation function to calculate hidden-layer output.
5. Pass the hidden output to the output layer.
6. Calculate the predicted output.
7. Repeat the process for the given training data.
8. Display the predicted results.

Flowchart



Code

```
import math
```

```
# Sigmoid activation function
```

```
def sigmoid(x):
```

```
    return 1 / (1 + math.exp(-x))
```

```
# Input data
```

```
inputs = [
```

```
    [0, 0],
```

```
    [0, 1],
```

```
    [1, 0],
```

```
    [1, 1]
```

```
]
```

```
# Expected output (AND operation)
```

```
expected = [0, 0, 0, 1]
```

```
# Weights and bias
```

```
w1 = 0.5
```

```
w2 = 0.5
```

```
bias = -0.7
```

```
print("Feed Forward Neural Network Output:")
```

```
for x in inputs:
```

```
    # Hidden/Output neuron calculation
```

```
    z = (x[0] * w1) + (x[1] * w2) + bias
```

```
    # Activation
```

```
    output = sigmoid(z)
```

```
    # Classification
```

```
    if output >= 0.5:
```

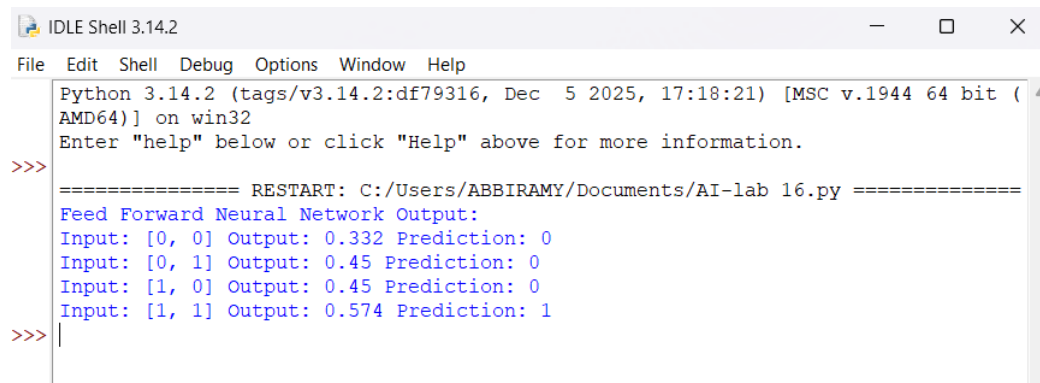
```
        prediction = 1
```

```
    else:
```

```
        prediction = 0
```

```
print("Input:", x,  
      "Output:", round(output, 3),  
      "Prediction:", prediction)
```

Output



```
IDLE Shell 3.14.2  
File Edit Shell Debug Options Window Help  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>> ===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 16.py =====  
Feed Forward Neural Network Output:  
Input: [0, 0] Output: 0.332 Prediction: 0  
Input: [0, 1] Output: 0.45 Prediction: 0  
Input: [1, 0] Output: 0.45 Prediction: 0  
Input: [1, 1] Output: 0.574 Prediction: 1  
>>> |
```

Result

The Python program successfully implemented a simple Feed Forward Neural Network and produced the expected classification results for the given input data.

Conclusion

The Feed Forward Neural Network processes input data in one direction through its layers to produce an output. It forms the basic structure of many neural-network-based classification and prediction systems.